ELECTROMECHANICAL SYSTEMS ENGINEERING & EMBEDDED IOT

Self-Powered Autonomous Hydro-Valve System

Energy-Harvesting Micro-Turbine Architecture & Sub-GHz LoRa Subterranean Telemetry

Focus: Embedded Controls & Energy Harvesting

Actuation: 12V High-Torque Motorized Valve

RF Protocol: Sub-GHz LoRa (Low Quiescent Sleep)

EXECUTIVE SUMMARY

This project features the development of an autonomous, energy-surplus smart valve system designed to eliminate the need for expensive underground conduit and power trenching in agricultural and irrigation environments. Standard subterranean valves require hardwired power grids or recurring battery servicing. By coupling an in-line micro-hydro generator directly to the fluid flow path, this architecture harvests ambient hydraulic kinetic energy during normal operation, establishing a perpetual net-positive battery surplus. The system utilizes sub-GHz LoRa wireless communication to achieve reliable signal penetration through soil and valve pits while consuming orders of magnitude less power than conventional Wi-Fi or Bluetooth.

TECHNICAL SPECIFICATIONS & SUBSYSTEM CONSTRAINTS

PARAMETER	TARGET CONSTRAINT	ACHIEVED PERFORMANCE	STATUS
Power Topology	Zero external utility wiring	100% In-line hydro-harvesting	Target Met
Energy Balance	Perpetual net-positive balance	Power surplus / battery maintenance-free	Target Met
RF Propagation	Subterranean transmission capability	Sub-GHz LoRa (Heltec ESP32-S3 V4)	Target Met
Quiescent Drain	Ultra-low deep sleep baseline	CAD sleep cycling (microamps)	Target Met
Actuation Torque	12V DC full-flow motorized ball valve	TPS61088 synchronous 12V boost rail	Target Met
Reverse Protection	Zero battery backfeed during stagnation	1N5819 Schottky isolation barrier	Target Met

THEORETICAL ELECTRICAL & FLUIDIC ARCHITECTURE

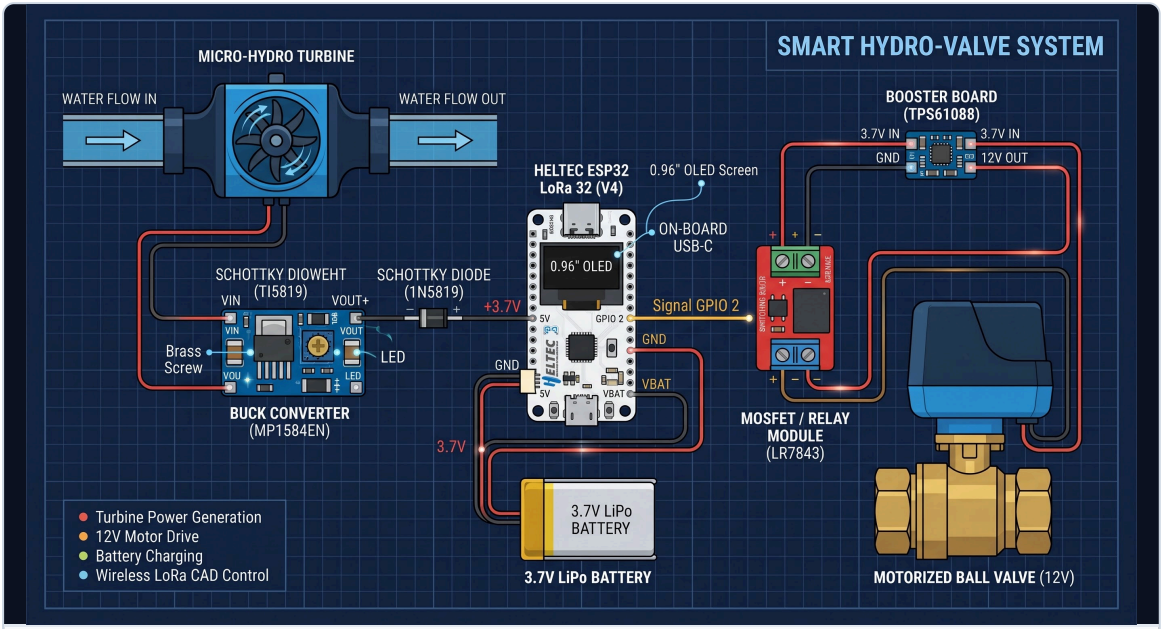


Figure 1: Complete System Schematic. Micro-hydro generator output is rectified and regulated via an MP1584EN buck converter to replenish the 3.7V LiPo cell. A high-efficiency TPS61088 synchronous booster steps up battery voltage to drive the 12V motorized ball valve via an LR7843 MOSFET switch, coordinated by a Heltec ESP32 LoRa 32 V4 controller.

HARDWARE SUBSYSTEMS & FUNCTIONAL DESIGN

1. Kinetic Energy Harvesting

An in-line Pelton-style DC micro-turbine captures kinetic energy directly from fluid discharge. Variable DC generation is routed through an adjustable MP1584EN step-down buck converter calibrated for safe 3.7V LiPo charging, protected with a 1N5819 low-drop Schottky diode against reverse battery leakage during no-flow intervals.

2. Dual-Rail Power Management

A high-capacity 3.7V 3000mAh LiPo battery acts as the system energy reservoir, decoupling motor burst demand from real-time generation. A dedicated TPS61088 synchronous boost converter elevates the 3.7V rail to a robust 12V DC, supplying sufficient inrush amperage for motorized valve transit without brownout.

3. Wireless Sub-GHz Telemetry

The system controller utilizes a Heltec ESP32 LoRa 32 (V4) module featuring an on-board 0.96" OLED diagnostic display. Operating across sub-GHz frequencies bypasses the steep attenuation common to 2.4 GHz Wi-Fi and Bluetooth when transmitting through dense soil, wet turf, and submerged composite utility boxes.

4. Low-Loss Valve Actuation

Actuation is executed via a solid-state LR7843 MOSFET module triggered directly from GPIO 2. A 12V motorized ball valve was chosen over solenoid valves to completely eliminate continuous holding current and eradicate disruptive hydraulic water hammer upon closure.

BENCHTOP CIRCUITRY & HARNESS INTEGRATION

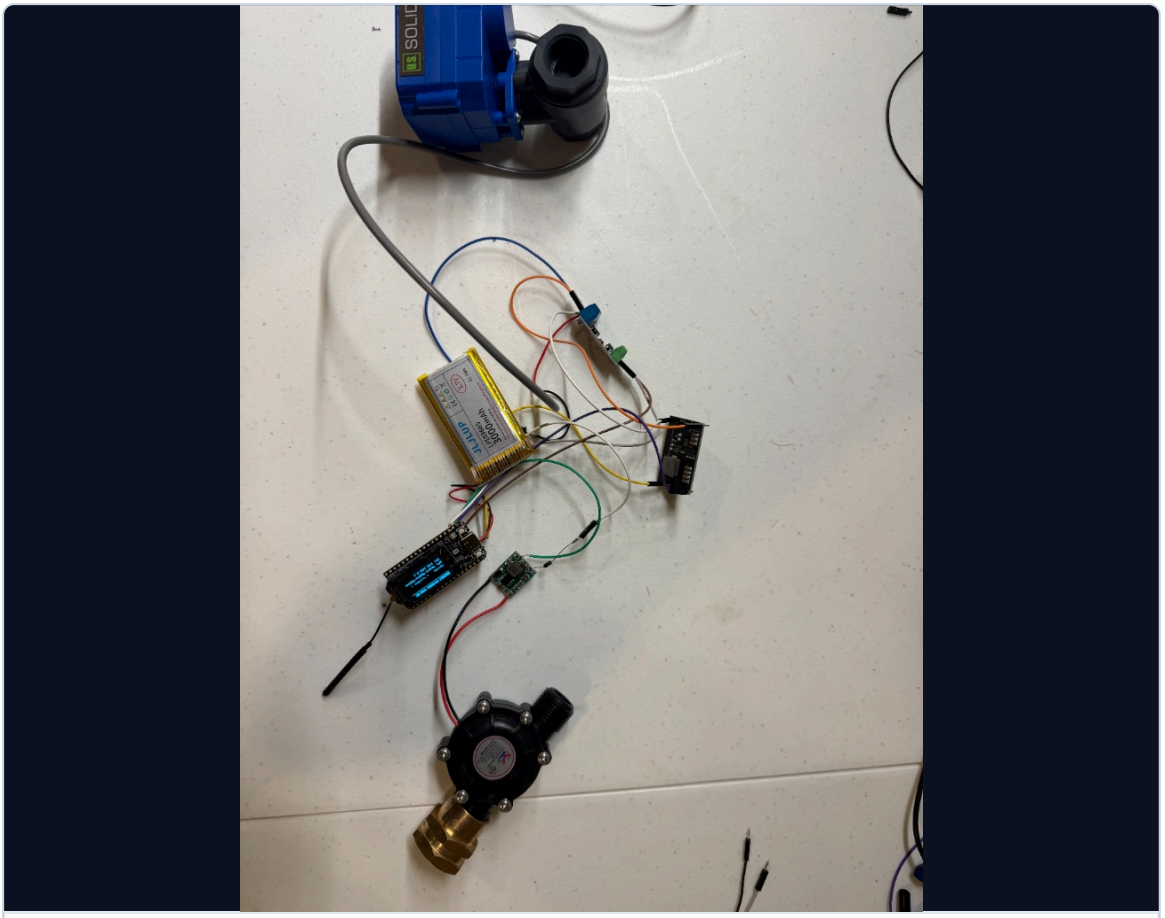


Figure 2: Electromechanical Benchtop Prototype. Validation setup showing interconnected power conditioning circuitry, 3000mAh LiPo storage, synchronous boost stage, Heltec ESP32 LoRa microcontroller running telemetry logic, DC micro-hydro generator, and the U.S. Solid motorized ball valve.

MECHANICAL PACKAGING & FIELD ENCLOSURE

Subterranean valve environments present extreme humidity, soil exposure, and moisture ingress challenges. The electronics suite, battery cell, and step-up modules are packaged inside a custom precision enclosure (Curatech Outdoor Systems). The enclosure features internal isolated mounting planes to decouple power-converter thermals from sensitive microcontroller logic, recessed perimeter screw anchors for environmental sealing, and direct inline brass union coupling to the turbine and valve bodies.



Figure 3: Fully Packaged Field-Ready System. Complete prototype assembly showing the sealed protective electronics chassis directly coupled to the in-line micro-hydro generator inlet and motorized valve outlet, providing a true wire-free drop-in plumbing replacement.

ENGINEERING ITERATIONS & PROBLEM SOLVING

- **Energy Surplus & Deep-Sleep Scheduling:** Continuous polling depleted battery reserves prematurely. Engineered a low-power scheduling loop utilizing LoRa Channel Activity Detection (CAD). The ESP32 remains in deep sleep (~15 µA) until pinged, ensuring standard irrigation flow intervals generate a substantial net-positive energy surplus.
- **High Inrush Motor Surge Mitigation:** Valve transit initiates with heavy inductive inrush that initially caused 3.7V logic brownout. Resolved by sizing a high-efficiency TPS61088 synchronous step-up converter with low-ESR ceramic filtering to cushion peak current draws up to 2.5A without disrupting logic rails.
- **Reverse Leakage Current Elimination:** When fluid stopped flowing, unpowered buck regulation circuitry acted as a passive parasitic load on the LiPo cell. Integrated a 1N5819 Schottky barrier diode on the charging rail, bringing quiescent back-leakage to negligible microamp levels.